

Xingji Yan

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Software Engineer (Java + React/TypeScript) who owns multi-tenant product end-to-end: schema, APIs, and UIs used by **20+ organizations / 500+ concurrent users**, then **on-call** when it breaks. Shipped a production **AI action layer** (MCP tools, JSON Schema, guardrails). Daily **Claude Code / Cursor / Copilot** — **reviews AI-generated code** before merge. **M.S., Carnegie Mellon.**

SKILLS

Languages: Java, TypeScript, JavaScript (ES6+), SQL, Python, Go, C#; Kotlin (personal projects)

Backend: Spring MVC/Boot, Hibernate/JPA, REST APIs, WildFly/JBoss, event-driven & batch processing, workflow orchestration, multi-tenant SaaS, OAuth 2.1; Gradle & Maven, JUnit

Frontend: React, TypeScript, Next.js, Vue.js, HTML5/CSS3, responsive UI, drag-and-drop editors, reusable components, product-minded interface design

Data: Oracle, PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, complex SQL & query optimization, stored procedures, Snowflake, Apache Kafka, data pipelines & asynchronous worker pools

AI-assisted & agentic development: **Claude Code, Cursor, GitHub Copilot** daily; agentic coding workflows, spec-driven development, rapid prototyping;

evaluate and reject unsafe AI output before merge; production MCP / JSON Schema, AI guardrails; IBM RAG & Agentic AI certified

Cloud & quality: AWS, Docker, Kubernetes, Jenkins CI/CD, GitHub Actions, Terraform/CDK, Datadog, Linux, SonarQube, JMeter; GCP Professional Cloud Architect certified

PROFESSIONAL EXPERIENCE

JASCI Software — Software Engineer

Melbourne, FL | Feb 2023 — Jul 2026

Smart Warehouse Management System (V1 → V2 AI-enabled platform) — multi-tenant SaaS serving 20+ 3PL customers and 500+ concurrent users; owned Java/Spring services and React/TypeScript UIs from requirements through production on-call

Product Owned End-to-End

- Built a configurable **3PL billing engine** (pricing rules, billing cycles, multi-currency, payment terms) automating invoice generation end-to-end with **idempotent billing reruns, currency-precision handling, and reconciliation checks against source transactions** — replacing **~10 hours of manual reconciliation per customer each month**.
- Built **React/TypeScript** consoles that streamline business counterparts' workflows (billing configuration, inventory operations) — responsive UIs used daily by **500+ concurrent users**, each owned end-to-end from **schema and API through interaction detail**.
- Built the **drag-and-drop workflow designer** in React/TypeScript: drag state management, **live dependency validation surfaced inline**, and reusable configuration components shared across modules — letting non-engineers author **1,000+ step processes without a release**.

Reliability Under Load

- Built event and batch processing pipelines for peak load with asynchronous architecture: moved long-running jobs (order ingestion, wave allocation) **off request threads onto async worker pools** and tuned **Oracle UCP connection pools** — eliminating recurring pool-exhaustion outages and keeping API latency stable through seasonal volume spikes.
- Extended the platform's **workflow orchestration engine** with dependency-aware step validation and **safe mid-process recovery**, making long-running multi-step processes **retry-safe and resumable without data loss** — turning process changes from full release cycles into **same-day configuration**.
- Cut inventory search latency from 10+ seconds to sub-second** on million-record inventory tables: root-caused **N+1** query bottlenecks and redesigned SQL and DAO access patterns with in-memory de-duplication — reducing database round trips by **~95%** and keeping it sub-second as tenant inventories grew.
- Carried **production on-call rotation across 20+ customer sites** and shifted incident response from reactive to preventive: root-caused the most frequent failure classes (**Oracle lock contention, malformed partner interface data**), added monitoring and targeted logging, and established team-wide error-classification, failure-mode test cases, and pre-release verification practices — cutting typical recovery **from days to hours**.

AI in Production, With a Bar

- Took a new **AI action layer from concept to production (V2)**: defined the technical plan and exposed **10+ core operations** (inventory lookup, wave release, reporting) as **MCP tools with JSON Schema contracts** and audit logging — collapsing multi-screen workflows into single natural-language requests and cutting routine operations from **minutes to seconds**.
- Engineered **guardrails for AI over live warehouse data**: strict schema validation **rejecting malformed tool calls before execution, idempotent write handlers** for retry-safe operations, and structured audit trails for every AI-initiated action.
- Introduced **agentic development workflows** to the team: drove daily implementation with **Claude Code and Cursor** from structured specs, **reviewed, tested, and rejected unsafe AI-generated code** before merge, then **codified the repeatable patterns into team spec and code-review standards** and led adoption through technical discussions.

One Model, Many Customers

- Partnered with product in agile delivery and with **20+ customer organizations** to converge conflicting requirements into one extensible model: normalized per-client customization into **metadata-driven configuration** (general-code tables, configurable workflows) instead of client-specific forks — cutting new-customer onboarding **from weeks of code forking and redeployment to days of configuration**.
- Implemented **tenant-scoped OAuth 2.1** authorization and propagated tenant, company, and fulfillment-center context through the service and data-access layers — enforcing **data isolation for 20+ 3PL customers on shared infrastructure**.

Sonos Inc — Software Engineer Intern, Universal Search

Remote | Jun 2022 — Aug 2022

- Built a **real-time keyword search pipeline**: streamed database changes into the search index incrementally via **Kafka** and cached weekly top queries in an in-memory **Redis** store — improving search speed by **30%**.
- Automated frontend delivery and code-quality gates: deployed a **React application via AWS S3/CloudFront** and integrated **SonarQube** into the Android CI pipeline (**Jenkins, Kubernetes, Gradle**) — blocking quality regressions before merge.

Tencent — Software Engineer Assistant, WeChat Applet (Smart Property)

Remote | Oct 2020 — Dec 2020

- Built **Spring Boot services and REST APIs** for user and payment workflows; introduced **Redis** caching to cut database load by **30%** and containerized services with **Docker**.
- Load-tested payment and user APIs with **JMeter** under **8,000-concurrent-user** scenarios; profiled and fixed the top bottlenecks, reducing **P99 latency by 30%** under peak load.

PROJECTS

- Clearbay** ([live](#) / [repo](#)) — **Java 21 / Spring**: tenant JWT, billing, MCP guardrails, JUnit. **Dockline** ([live](#) / [repo](#)) — 40-case eval harness that scores AI tool traces instead of trusting them. **Durable Brief** ([live](#) / [repo](#)) — **Next.js** briefing desk: parallel research, draft, evaluator-optimizer loop.

EDUCATION

Carnegie Mellon University — M.S. in Electrical and Computer Engineering

Pittsburgh, PA | Dec 2022

Florida Institute of Technology — B.S. in Electrical and Computer Engineering

Melbourne, FL | May 2021

PATENT & CERTIFICATIONS

Certifications: IBM RAG and Agentic AI; Duke MLOps | Machine Learning Operations; Advanced ML on Google Cloud; Google Cloud Professional Cloud Architect.

Patent: CN205178942U (2016), flexible solar panel field emergency power supply.